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Grouping techniques for large-scale cluster dynamics simulations of reaction diffusion processes

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Abstract

Cluster dynamics is a powerful, high fidelity, mesoscale method for modeling the kinetic evolution of point defects, impurities, and their clusters in materials and is commonly used in studying radiation damage. These methods excel at modeling nucleation, but often require too many equations to successfully model the long term growth and coarsening that govern microstructural evolution. One solution to this problem is to group equations into a coarser approximation of the cluster size distribution function which can reduce the cost of solution by many orders of magnitude. While such grouping methods have been advanced for a limited class of problems, no reliable method currently exists for the general case. This paper advances a framework for grouping arbitrary cluster dynamics problems, and develops several competing schemes based on that framework. These schemes are each evaluated against a variety of test problems designed to assess their accuracy, robustness, and efficiency.

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(Some figures may appear in colour only in the online journal)

1. Introduction

Cluster dynamics methods are a powerful rate theory approach developed to simulate and analyze the kinetics of evolving material microstructures. These models are particularly useful

in studying radiation damage, where they have been applied to processes as varied as noble gas behavior and bubble formation [1–3], dislocation loop nucleation [4–7], and radiation induced and enhanced precipitation [8, 9] with potential applications to radiation creep and solute segregation. These methods rely on evolving a set of coupled—and often spatially dependent—differential equations through time. Each equation represents the concentration of a particular defect or solute species and takes the form

$$\frac{dC_i}{dt} = D_i \nabla^2 C_i - D_i k_i^2 C_i + R_i(\mathbf{C}) + g_i, \quad (1)$$

where C_i is the concentration of the i th species, D_i is its diffusivity, k_i^2 is the net sink strength for its removal, g_i is its source rate, and $R_i(\mathbf{C})$ is the rate of consumption or production through interaction with other species. The equations are coupled through these reaction rates. As an example, one might construct a problem where each C_i represents the population of precipitates containing i atoms and the reaction terms describe the depletion of smaller clusters and appearance of larger ones as solutes arrive at the existing precipitate distribution.

These methods provide a substantial improvement over the ‘classical rate theory’ approach in that they compute the full size distribution of irradiation induced damage features rather than a single average population. A major advantage of this approach is that the nucleation phase of damage formation can be treated consistently with growth, where older methods must assume feature densities or alternative nucleation models [10–12]. These methods, however, must overcome a significant technical hurdle for broad application. As posed, the natural resolution for cluster dynamics is of the atomic scale, but the desired scale to model engineering relevant microstructural changes involves defects or features with sizes of nanometers to hundreds of nanometers. As an example, it requires on the order of 10^8 equations to simulate the formation of voids a few tens of nanometers in diameter. This may prove manageable given a limited number of reaction pathways, but if the number of constituents increases, as for modeling gas bubbles or complex precipitates, the situation becomes completely infeasible.

This combinatorial explosion has typically limited the application of cluster dynamics to the description of small, simple systems, and for larger or more complex systems researchers have turned to other methods. Kinetic Monte Carlo (kMC) [13, 14] and stochastic cluster dynamics [15–17] methods both offer an attractive alternative to the deterministic approach. These methods do not suffer from the combinatorial explosion from large or multi-component cluster spaces in the same way that deterministic approaches often do. These methods offer a number of additional potential advantages as well. For instance, object kMC models do not require the rate constants embedded in cluster dynamics which are computed from mean field conditions and may be incorrect under certain circumstances such as during spatially correlated damage events. While moving to stochastic approaches makes a solution feasible, the cost associated with attaining damage doses relevant to major microstructural changes is quite high and most such simulations reach terminal doses of a few displacements per atom (dpa) or less. These methods have not yet demonstrated the capability to model the microstructural changes in fuels or in structural materials for fast reactors which must endure doses of hundreds of dpa or more.

On the deterministic side it has long been noted that the Fokker–Plank equation can be used to represent the size distribution and its evolution, however a general application remains elusive [18–20]. The most successful effort to date shows significant and exciting improvement for specific cases but requires a transition regime between a region of discrete treatment at small sizes and Fokker–Plank at larger sizes [20], and it remains unclear how reliable this is or if it can be formulated generally for broad application. The final alternative coarsens the

description of the size distribution in a cluster dynamics approach by simply neglecting many of the individual species which share similar properties with their neighbors. Rather, several species or even several thousand species might be represented by a single equation instead. This is more complicated than it might appear. Due to the difficulty in properly formulating the reaction terms between the remaining equations, early attempts at such a ‘grouping scheme’ failed. To date, a grouping scheme based on conserving multiple moments of the cluster distribution has demonstrated a significant advantage in reliability over previous attempts [21].

However, the method of Golubov and co-workers only allows for the mobility of monomer species as originally developed, and thereby growth or shrinkage of clusters only through the monomer flux. It is now well-established that in radiation damage problems much of the primary damage is created in the form of point defect clusters [22–24], and that those clusters may have significant mobility providing additional mechanisms for reaction as observed experimentally [25–27], and predicted theoretically [28, 29]. Though the moment scheme as is can be massaged to accommodate a few extra mechanisms, the complexity grows rapidly. It is desirable to find a generally formulated grouping scheme, which can accommodate arbitrary reactions between arbitrary clusters regardless of their positioning in the size distribution, not just to handle primary damage, but to open up the possibility of modeling microstructural developments such as the coalescence, loss to sinks, and spatial transport of larger clusters.

This paper constructs several grouping schemes which meet this requirement, including both novel approaches and improvements to existing methods. First, we develop a family of grouping schemes based on interpolation of the cluster distribution. Secondly, we apply a similar analysis to develop a generalization of the moment conserving grouping method which allows for arbitrary reaction networks as opposed to only a monomer flux. The resulting schemes are tested against a series of test problems designed to span several phenomenon of interest in radiation damage.

2. Methods

The evolution of features in cluster dynamics is heavily dependent on the coupling terms, R in equation (1). For radiation damage problems, the full description for a given R_i follows

$$R_i(\mathbf{C}) = \sum_{n+m \rightarrow i} k_{n,m}^+ C_n C_m - \sum_n k_{i,n}^+ C_i C_n + k_{i+1}^- C_{i+1} - k_i^- C_i, \quad (2)$$

where the first term describes the creation of cluster i through the reaction of two parent clusters, the second term involves the loss of cluster i in reactions with other clusters and the final two terms describe the production or loss of a cluster through emission of one of its components. In principle, additional emission pathways are conceivable in addition to the monomer mechanism assumed here but in practice these dissociation terms are far less numerous than the reaction terms. The k^+ and k^- represent rate coefficients for reaction and dissociation respectively. For instance, in a problem with only an interstitial and vacancy monomer flux, the reaction network can be written

$$R_i(\mathbf{C}) = (k_{1,i-1}^+ C_{i-1} C_1 + k_{-1,i+1}^+ C_{i+1} C_{-1}) - (k_{1,i}^+ C_i C_1 + k_{-1,i}^+ C_i C_{-1}) + k_{i+1}^- C_{i+1} - k_i^- C_i \quad (3)$$

but more complex problems will include additional terms for each reacting species. Monomer equations require additional terms for the monomers emitted from the entire distribution.

2.1. Constructing groups by interpolation

Grouping methods aim to choose some smaller subset of $\mathbf{C}$, thus reducing the number of R values which must be computed at each integration step. For complex problems with many reacting species, this also potentially reduces the number of terms required to compute each R . We denote this subset as $\hat{\mathbf{C}}$. The challenge in constructing a valid group scheme lies in determining a consistent means of computing a given R using only the values in $\hat{\mathbf{C}}$ when the natural formulation in equation (2) requires, in principle, the entire set $\mathbf{C}$. The general form of the reaction network now reads

$$R_i(\hat{\mathbf{C}}) = \sum_{p,q} a_{p,q,i}^+ \hat{C}_p \hat{C}_q - \sum_p b_{i,p}^+ \hat{C}_i \hat{C}_p + \sum_p a_{i,p}^- \hat{C}_p - b_i^- \hat{C}_i, \quad (4)$$

where the means of determining the coefficients a and b from the reaction rate constants k is unclear. The remainder of this section lays out a framework for constructing such coefficients.

It will be helpful to introduce the size distribution function (SDF) as $C(x)$ where $C_i = C(i)$. In practice, the SDF can be approximated by a low order piecewise equation defined by the subset of points $\hat{\mathbf{C}}$. The unknown elements of $\mathbf{C}$ required for equation (2) can be approximated from the interpolating functions. Pursuing such computations directly at each timestep becomes expensive, however, and a derivation for the coefficients a and b to render equation (4) solvable is preferred.

Constructing these coefficients requires introducing a set of basis functions $v(x)$ such that the SDF approximation can be written

$$C(x) = \sum_i \hat{C}_i v_i(x), \quad (5)$$

where the basis functions are chosen to construct the desired piecewise function. A wide variety of interpolation strategies can be imagined, each with its own set of basis functions. Inserting this definition of the SDF directly in equation (2) gives

$$R_i(\hat{\mathbf{C}}) = \sum_{n+m \rightarrow i} \left[k_{n,m}^+ \sum_p v_p(n) \hat{C}_p \sum_q v_q(m) \hat{C}_q \right] - \sum_n \left[k_{i,n}^+ \hat{C}_i \sum_p v_p(n) \hat{C}_p \right] + k_{i+1}^- \sum_p v_p(n) \hat{C}_p - k_i^- \hat{C}_i \quad (6)$$

which has the same form of equation (4) and reveals how to construct coefficients for the grouped network. Namely,

$$a_{p,q,i}^+ = \sum_{n+m \rightarrow i} (k_{n,m}^+ v_p(n) v_q(m)), \quad (7a)$$

$$b_{i,p}^+ = \sum_n (k_{n,i}^+ v_p(n)), \quad (7b)$$

$$a_{i,p}^- = \sum_{n+m \rightarrow i} (k_p^- v_p(n)), \text{ and} \quad (7c)$$

$$b_i^- = k_i^- \quad (7d)$$

which are valid regardless of the form of the basis function and are independent of the actual concentration values so they need be computed only once, before time integration begins. Enforcing constant cluster properties across each group allows the rate constants to be extracted from the summations such that they can be converted to analytical expressions and the grouped coefficients can be computed inexpensively.

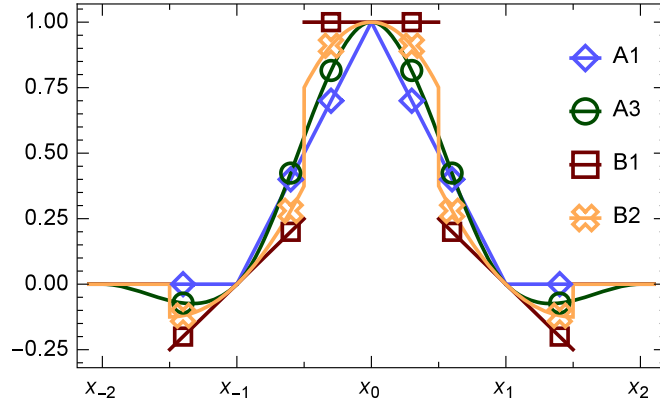


Figure 1. Basis functions for the four interpolation schemes examined in this paper as labeled in table 1.

Any piecewise polynomial interpolation which can be decomposed into basis functions can be used to construct a grouping scheme through this framework. In this paper, we will limit our analysis to four possible interpolating functions. These functions are listed in table 1 and the basis functions corresponding to them are plotted in figure 1. In some of these schemes, piecewise segments of the SDF are defined such that the domain of each component spans the space between two nodes. Two such schemes are used here, one of which is simple linear interpolation and the other is a third order spline (the A type schemes in table 1). Alternatively, each component function might be defined to span some domain $x_i^{\min}$ to $x_i^{\max}$ such that x_i lies in the center of the domain. We refer to these as B type schemes, and have examined a first order scheme based on the secant method and a second order scheme based on the Lagrange interpolating polynomial. It is worth mention that the SDF and its derivatives are not necessarily continuous when transitioning from one piece to the next, and such discontinuities and cusps are evident in the basis functions where they occur.

2.2. Constructing groups by conserving moments

The moment conserving approach suggested by Golubov *et al* [21] offers a compelling alternative to the SDF interpolation approach outlined above. However, as mentioned in the introduction, that scheme as proposed was formulated only for monomer fluxes and is poorly suited for application to arbitrary cluster dynamics problems. In this section, we will outline a means to formulate the moment conserving scheme in terms of basis functions—as was done for SDF interpolations in the previous section—to allow for reactions between groups rather than solely monomers. In the moment conserving scheme, the SDF is described by two equations for each group such that the magnitude and first derivative of the distribution are described in separate variables. To approximate the SDF in the i th group, one can write

$$C(x) = L_i^0 + \frac{x - x_i^0}{1 + \Delta x_i} L_i^1 \quad (8)$$

for a group containing clusters of sizes $x_i^0 - \Delta x_i$ to $x_i^0 + \Delta x_i$. The denominator in the second term is not present in the original formulation of this scheme, but is included here to avoid losses of precision in L^1 as group widths become large.

Table 1. List of interpolation schemes and the basis functions associated with them. Identifying labels are provided for each scheme denoting domain class and order of the interpolating functions.

Scheme	Label	Basis functions $v_i(x)$	
Linear	A1	$\frac{x - x_{i-1}}{x_i - x_{i-1}}$	if $x > x_{i-1}$ and $x \leq x_i$
		$\frac{x_{i+1} - x}{x_{i+1} - x_i}$	if $x > x_i$ and $x < x_{i+1}$
Secant	B1	0	else
		$\frac{x - x_{i-1}}{x_i - x_{i-2}}$	if $x \geq x_{i-1}^{\min}$ and $x \leq x_{i-1}^{\max}$
		1	if $x \geq x_i^{\min}$ and $x \leq x_i^{\max}$
		$\frac{x_{i+1} - x}{x_{i+2} - x_i}$	if $x \geq x_{i+1}^{\min}$ and $x \leq x_{i+1}^{\max}$
Lagrange	B2	0	else
		$\frac{(x - x_{i-1})(x - x_{i-2})}{(x_i - x_{i-1})(x_i - x_{i-2})}$	if $x \geq x_{i-1}^{\min}$ and $x \leq x_{i-1}^{\max}$
		$\frac{(x - x_{i-1})(x - x_{i+1})}{(x_{i+1} - x_i)(x_{i+1} - x_i)}$	if $x \geq x_i^{\min}$ and $x \leq x_i^{\max}$
		$\frac{(x_{i+1} - x_i)(x_{i+2} - x)}{(x_i - x_{i+1})(x_{i+2} - x_i)}$	if $x \geq x_{i+1}^{\min}$ and $x \leq x_{i+1}^{\max}$
Cubic spline	A3	0	else
		$\frac{(x - x_{i-2})^2(x - x_{i-1})}{2(x_{i-2} - x_{i-1})^2(x_i - x_{i-1})}$	if $x > x_{i-2}$ and $x \leq x_{i-1}$
		$f_2(x)$	if $x > x_{i-1}$ and $x \leq x_i$
		$f_3(x)$	if $x > x_i$ and $x \leq x_{i+1}$
		$\frac{(x - x_{i+1})(x - x_{i+2})^2}{2(x_i - x_{i+1})(x_{i+1} - x_{i+2})^2}$	if $x > x_{i+1}$ and $x < x_{i+2}$
		0	else
$f_2(x) = \frac{(2x + x_{i-1} - 3x_i)(x - x_{i-1})^2}{(x_{i-1} - x_i)^3} - \frac{(x - x_i)(x - x_{i-1})^2}{2(x_{i-1} - x_i)^3}$ $+ \frac{(x - x_i)(x - x_{i-1})^2}{2(x_{i-1} - x_i)^2(x_i - x_{i+1})} - \frac{(x - x_i)^2(x - x_{i-1})}{2(x_{i-1} - x_i)^3}$ $f_3(x) = \frac{(x - x_{i+1})(x - x_i)^2}{2(x_i - x_{i+1})^3} - \frac{(x - x_{i+1})^2(x - x_i)}{2(x_{i-1} - x_i)(x_i - x_{i+1})^2}$ $+ \frac{(x - x_{i+1})^2(x - x_i)}{2(x_i - x_{i+1})^3} - \frac{(x - x_{i+1})^2(2x - 3x_i + x_{i+1})}{(x_i - x_{i+1})^3}$			

The appeal of this approach is that the number density of clusters can be accurately represented by L^0 , while adjustments to L^1 allow precise conservation of mass. If we define $N = \sum C(x)$ and $M = \sum xC(x)$ then for the i th group

$$N_i = \sum_{x=x_i^0-\Delta x_i}^{x_i^0+\Delta x_i} L_i^0 + \frac{x - x_i^0}{1 + \Delta x_i} L_i^1 \text{ and} \quad (9a)$$

$$M_i = \sum_{x=x_i^0-\Delta x_i}^{x_i^0+\Delta x_i} xL_i^0 + x \frac{x - x_i^0}{1 + \Delta x_i} L_i^1 \quad (9b)$$

which simplify to

$$N_i = (1 + 2\Delta x_i)L_i^0 \text{ and} \quad (10a)$$

$$M_i = (1 + 2\Delta x_i) \left(x_i^0 L_i^0 + \frac{\Delta x_i}{3} L_i^1 \right) \quad (10b)$$

respectively. Consequently, changes in L^0 and L^1 for an SDF component can be determined from changes in the mass and number density in the corresponding group according to

$$\Delta L_i^0 = \frac{1}{1 + 2\Delta x_i} \Delta N_i \text{ and} \quad (11)$$

$$\Delta L_i^1 = \frac{3}{\Delta x_i(1 + 2\Delta x_i)} (\Delta M_i - x_i^0 \Delta N_i) \quad (12)$$

and in turn the change in number density for the i th group can be found by summing over constituent clusters as

$$\Delta N_i = \sum_{x=x_i^0-\Delta x}^{x_i^0+\Delta x} R_x(C) \quad (13)$$

and the change in net mass in the group as

$$\Delta M_i = \sum_{x=x_i^0-\Delta x}^{x_i^0+\Delta x} x R_x(C), \quad (14)$$

where $R(x)$ must be determined from the SDF. Defining basis functions as

$$v_i^0(x) = \begin{cases} 1 & \text{if } x_i^0 - \Delta x_i \leq x \leq x_i^0 + \Delta x_i \\ 0 & \text{else,} \end{cases} \quad (15)$$

$$v_i^1(x) = \begin{cases} \frac{x - x_i^0}{1 + \Delta x_i} & \text{if } x_i^0 - \Delta x_i \leq x \leq x_i^0 + \Delta x_i \\ 0 & \text{else} \end{cases} \quad (16)$$

for L^0 and L^1 respectively allows explicit construction of a grouped reaction network. Using equation (13) in (11) and writing the reaction network in terms of the moment basis functions gives

$$\begin{aligned} R_i^0(\mathbf{L}) = & \frac{1}{1 + 2\Delta x_i} \left(\sum_{n+m \rightarrow x \in i} \left[k_{n,m}^+ \sum_p (v_p^0(n) L_p^0 + v_p^1(n) L_p^1) \right. \right. \\ & \times \left. \sum_q (v_q^0(m) L_q^0 + v_q^1(m) L_q^1) \right] \\ & - \sum_n \left[\sum_{x \in i} k_{x,n}^+ (v_i^0(x) L_i^0 + v_i^1(x) L_i^1) \times \sum_p (v_p^0(n) L_p^0 + v_p^1(n) L_p^1) \right] \\ & + \sum_{x \in i} [k_{x+1}^- (v_p^0(x+1) L_p^0 + v_p^1(x+1) L_p^1)] \\ & \left. - \sum_{x \in i} [k_x^- (v_p^0(x) L_p^0 + v_p^1(x) L_p^1)] \right) \end{aligned} \quad (17)$$

for L^0 values. Similarly using both equations (13) and (14) in (12) produces the grouped network

$$\begin{aligned}
R_i^1(\mathbf{L}) = & \frac{3}{\Delta x_i(1 + 2\Delta x_i)} \left(\sum_{n+m \rightarrow x \in i} \left[k_{n,m}^+ \sum_p (v_p^0(n)L_p^0 + v_p^1(n)L_p^1) \right. \right. \\
& \times \sum_q (v_q^0(m)L_q^0 + v_q^1(m)L_q^1) \times (n + m - x_i^0) \Big] \\
& - \sum_n \left[\sum_{x \in i} k_{x,n}^+ (v_i^0(x)L_i^0 + v_i^1(x)L_i^1) \right. \\
& \times \sum_p (v_p^0(n)L_p^0 + v_p^1(n)L_p^1) \times (x - x_i^0) \Big] \\
& + \sum_{x \in i} [k_{x+1}^- (v_p^0(x+1)L_p^0 + v_p^1(x+1)L_p^1) \times (x - x_i^0)] \\
& \left. - \sum_{x \in i} [k_x^- (v_p^0(x)L_p^0 + v_p^1(x)L_p^1) \times (x - x_i^0)] \right) \quad (18)
\end{aligned}$$

for the time derivatives of L^1 equations.

This scheme generates more coefficients than the interpolation approach. For example, each a^+ from the interpolation scheme is expanded to eight a^+ for the moment scheme, because both reactant equations and the daughter equation can be either of the L^0 or L^1 class. The equivalent form of equation (4) for the moment scheme is

$$\begin{aligned}
R_i^0(\mathbf{L}) = & \left[\sum_{p,q} a_{i,p,q}^{0,0,0+} L_p^0 L_q^0 + \sum_{p,q} a_{i,p,q}^{0,1,0+} L_p^1 L_q^0 + \sum_{p,q} a_{i,p,q}^{0,0,1+} L_p^0 L_q^1 + \sum_{p,q} a_{i,p,q}^{0,1,1+} L_p^1 L_q^1 \right] \\
& - \left[\sum_p b_{i,p}^{0,0,0+} L_i^0 L_p^0 + \sum_p b_{i,p}^{0,1,0+} L_i^1 L_p^0 \right. \\
& + \sum_p b_{i,p}^{0,0,1+} L_i^0 L_p^1 + \sum_p b_{i,p}^{0,1,1+} L_i^1 L_p^1 \Big] \\
& + \sum_p a_{i,p}^{0,0-} L_p^0 + \sum_p a_{i,p}^{0,1-} L_p^1 - b_i^{0,0-} L_i^0 - b_i^{0,1-} L_i^1 \quad (19)
\end{aligned}$$

for L^0 terms with an equation of the same form but separate coefficients for L^1 . Examining equations (17) and (18), it is apparent that

$$a_{p,q,i}^{r,s,0+} = \sum_{x \in in+m \rightarrow x} [k_{n,m}^+ v_p^r(n) v_q^s(m)] \text{ and} \quad (20a)$$

$$a_{p,q,i}^{r,s,1+} = \sum_{x \in in+m \rightarrow x} [k_{n,m}^+ v_p^r(n) v_q^s(m) (n + m - x_i^0)], \quad (20b)$$

where r and s can each take a value of 0 or 1. Similar expressions can be extracted for the remaining coefficients, which in the interest of brevity are not listed explicitly here.

The additional complexity associated with using moments instead of interpolation arises from the need to sum over rates throughout the entire group rather than simply a single point. Consequently, it is absolutely essential to find analytical expressions to avoid doing these summations explicitly. It is possible to compute the summations of equation (4) for interpolation schemes by brute force summation, but the double sums required for the moment scheme quickly become too expensive. It is possible to find appropriate expressions of the

group widths and centers, and a process to do so is provided in appendix A. The original formulation of this scheme for monomer fluxes can be considered a special case of these analytical forms.

The largest drawback to this scheme is that the number of equations required is doubled compared to the interpolation schemes. Furthermore, the number of operations to evaluate the reaction network is increased by a factor of four in the general case. Finally, the expense required to compute the grouping coefficients is greater, even when analytic expressions are found to replace the summations, as these expressions are quite complex. These disadvantages, however, can be quickly overcome should the scheme allow wider groups.

Lastly, we mention that all schemes can easily be extended to clustering problems with multiple components (e.g. noble gas bubbles) by defining a basis function in each component dimension. Each component lies on an orthogonal axis in phase space. Accordingly, summations can be performed independently along each component direction and then simply multiplied. For the moment scheme, an additional set of L equations must be included for each new component dimension. Thus the number of equations per group scales with the number of components as $n + 1$ and the number of terms in the reaction network per group as $(n + 1)^3$.

3. Results and discussion

3.1. Test problems

To probe the performance and reliability of the grouping schemes we have devised three test problems, each of which targets a different aspect of the physics commonly encountered in radiation damage problems. The first is a simple case of void swelling under Frenkel pair damage conditions with a monomer flux. This problem was included because it is simple enough and small enough to be solved with a cluster dynamics model that does not employ grouping. The second is a loop nucleation and growth problem, using a full cascade damage spectrum. Interstitial and vacancy clusters of arbitrary size are included, with the possibility of diffusion and coalescence for all interstitial clusters. A detailed description of this problem is available [4, 30], and it tests the ability of the schemes to handle dense reaction networks. Finally, we include a helium bubble formation problem, which will test the multi-component capability of these schemes. Table 2 summarizes the basic setup of these problems. All cases are taken to high doses, which will typically be inaccessible without a grouping scheme.

In each case the only contribution to the sink strength, k^2 in equation (1), is from network dislocations. A dislocation bias is applied to the sink strength for interstitial clusters in the form of a rate multiplier. This bias is also applied to reactions between interstitial clusters. The cluster diffusion, binding, and primary damage distributions are generally tuned to the expected properties of ferritic iron based alloys, and are listed in appendix B.

For each problem, group widths were selected by imposing a uniform spacing in physical size (cluster diameter) rather than the number of members. Vacancy clusters aggregate into spherical voids and interstitials into two-dimensional platelets so the group widths follow the cube of vacancy content and the square of interstitial content. Due to the assumption that rate constants are unchanging across a group, clusters whose properties differ radically from neighbors in phase space should not be grouped over. As applied to the radiation damage problems tested later in this paper, the smallest clusters fail to meet this criterion, as migration energies and sizes can change quite rapidly. Hence, we enforce a minimum size to the grouped region which varied from problem to problem but was typically around clusters with 20 members.

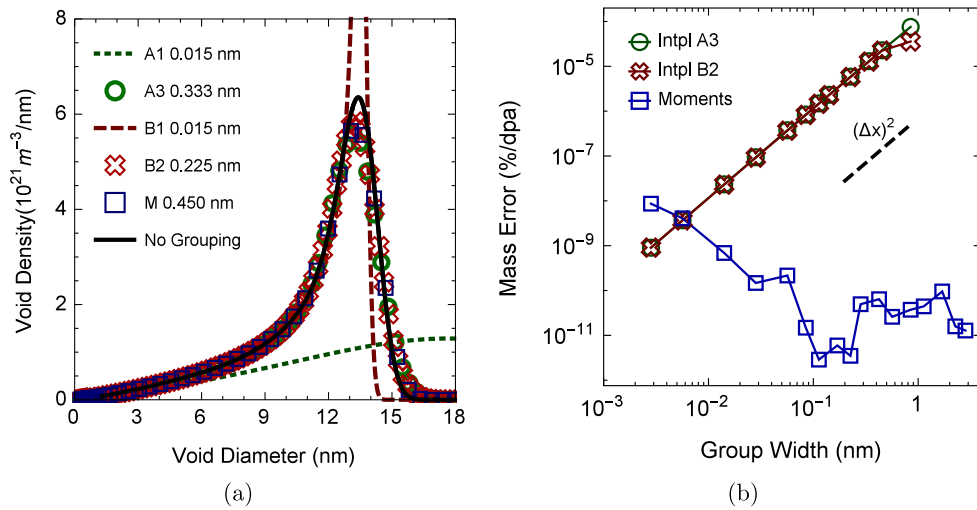


Figure 2. (a) Void size distribution for selected group widths and (b) system mass conservation as a function of group width at 100 dpa for the four interpolation schemes as well as the moment scheme when grouping is applied to the Frenkel pair test problem as compared to the non-grouped solution. The legend identifies interpolation schemes as labeled in table 1 or M for moment approach as well as the group width used.

Table 2. Test problem details. Cluster monikers v , i , and he refer to vacancy, interstitial, and helium content respectively.

Problem	Frenkel pair	Loops	Bubbles
Included clusters	$v_1 - v_{10^8}$ i_1	$v_1 - v_{10^4}$ $i_1 - i_{10^6}$	$v_1 - v_{10^8}$ i_1 $he_1 - he_{10^8}$
Mobile clusters	v_1, i_1	$v_{1-4}, \text{ all } i$	v_1, i_1, He_1
Source terms	v_1, i_1	$v_1 - v_9, i_1 - i_{20}$	$v_1 - v_9, i_1, he_1$
Dissociation	None	v_1, i_1	v_1, i_1, he_1
Dose rate (dpa s^{-1})	10^{-7}	10^{-3}	10^{-6}
Helium generation (appm dpa^{-1})	N/A	N/A	0.2
Terminal dose (dpa)	100	10	100
Temperature (C)	425	25	425
Dislocation density (m^{-2})	5×10^{13}	2.5×10^{14}	2.5×10^{14}
Dislocation bias (%)	5	5	1

3.2. Frenkel pair problem

Every interpolation scheme as well as the moment scheme were tested on the Frenkel pair problem. Additionally, the problem was run with a brute force cluster dynamics model with no grouping. This problem is the only one of the three where simulating to completion without grouping is feasible. Figure 2 shows a comparison between the void size distribution that develops by 100 dpa in each case, with void diameters calculated according to

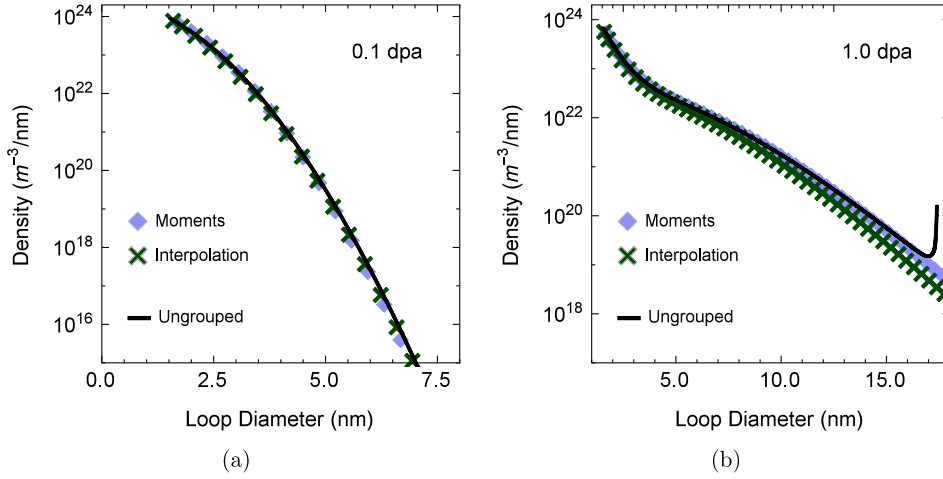


Figure 3. Comparison of low dose solutions from the discrete and grouped cluster dynamics simulations of the loop problem at (a) 0.1 dpa and (b) 1.0 dpa. Group widths of 0.38 nm were used for both the interpolation and moment based grouping schemes.

$$d = a_0 (3v/\pi)^{1/3}, \quad (21)$$

where v is the number of vacancies in a void. The moment scheme and interpolation schemes with order 2 or higher all perform reasonably well, while the lower order schemes have clearly diverged from the discrete solution even at comparatively small group widths. The type of error developed by the successful schemes is similar, with all showing a slight broadening of the SDF, while maintaining a correct peak location.

In the lower order methods, significant error has developed by 100 dpa, even with quite small group sizes. In the linear interpolation case, excessive growth occurred with mean sizes significantly larger than the correct solution. This spreading is qualitatively similar to the error observed in a method developed by Kiritani as an early attempt at grouping [31, 32]. Interestingly, the two schemes are functionally equivalent in a monomer flux, save for a few additional terms introduced at the interface between the grouped and ungrouped regions by Kiritani. Conversely, the secant method develops an excessive concentration, but with the peak size correctly placed. Due to the excessively poor performance of these schemes, neither were applied to the more complex problems.

These behaviors are both easily rationalized in terms of the Fokker–Planck equation, which illustrates that net growth and shrinkage are governed by the first derivative of the size distribution while spreading of the size distribution is governed by the second. In the linear interpolation method, the first derivative of the SDF has discontinuities at nodes so it is not surprising that the mean size diverges. In the secant approach the first derivative is continuous across the nodes, but constant, such that the second derivative is always zero. This prevents the size distribution from spreading such that sharp peaks cannot be relaxed. Consequently, in order to converge to the true solution, an interpolation scheme must be of at least second order with a derivative that is continuous across the nodes.

One measure of the robustness of each method is the degree to which it fails to conserve mass. Due to the nature of the moment method, it should be expected to conserve mass to numeric precision regardless of group size. The interpolation methods do not share this property and a mass leak can accumulate in the SDF over time. The mass error for several schemes is shown in figure 2(b). The mass excess accumulated throughout the simulation is

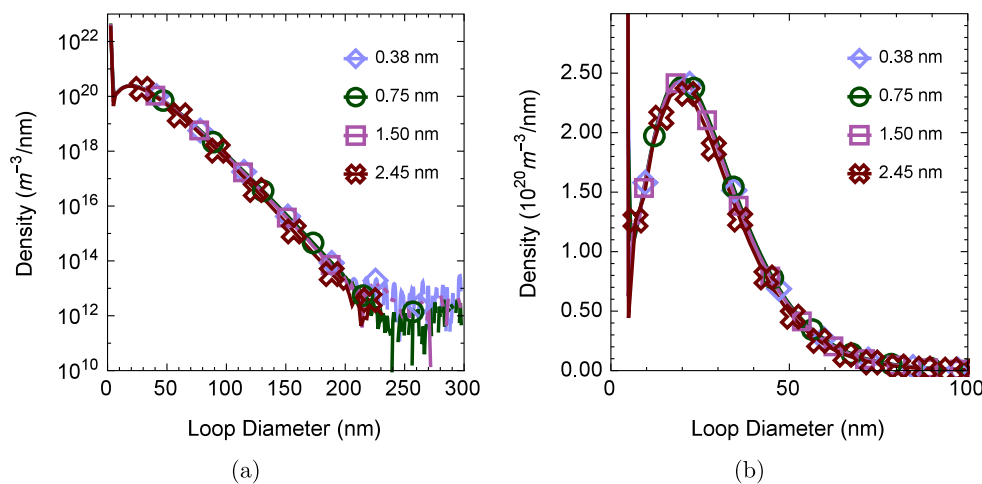


Figure 4. Loop size distributions at 10 dpa for moment grouping schemes for four group widths on (a) log and (b) linear scales. Occasional markers are added to demonstrate that all curves are present but indistinguishable.

surprisingly minimal for the interpolation methods. For both the second order Lagrange and cubic Hermite methods, the error converges with group width on the second order. For the moment method, the error hovers near numeric noise as expected, and the slight declining trend with group widths can be explained by the fewer equations required.

3.3. Loop problem

The loop problem, which involves the formation, diffusion, and aggregation of black dot damage produces a very dense reaction network as all of the interstitial clusters are deemed mobile and capable of coalescence regardless of size. Additionally, this problem accounts for the intrinsically one-dimensional migration of many interstitial clusters which follows a kinetics that is of third rather than second order in cluster concentration [33]. Based on the obvious failure of the first order interpolation methods on the simpler Frenkel pair problem, neither were applied in this case. The denser nature of the reaction networks requires implementation of analytic functions to compute the network coefficients, and doing so is complex. For this reason, only one of the two remaining interpolation schemes (the second order Lagrange) was applied to this problem and compared to the moment method.

The discrete cluster dynamics solution to this problem is only available at early doses when the features remain small. Figure 3 compares the distributions produced by the chosen interpolation scheme and the moment scheme to the discrete solution at times well below the dose target for this problem. At 0.1 dpa, both schemes match the distribution generated without a grouping scheme. By 1 dpa the three solutions begin to demonstrate noteworthy differences. In the ungrouped case, the distribution has approached the maximum allowable size, resulting in some accumulation of clusters at the cutoff point. At sizes below such corruption in the discrete calculations, however, the moment scheme closely follows the discrete distribution. For the interpolation approach, some deviation from the true solution has developed by this dose, and is readily visible in the tail to large sizes, though the distributions still align in the small size regime of 2–4 nm defects.

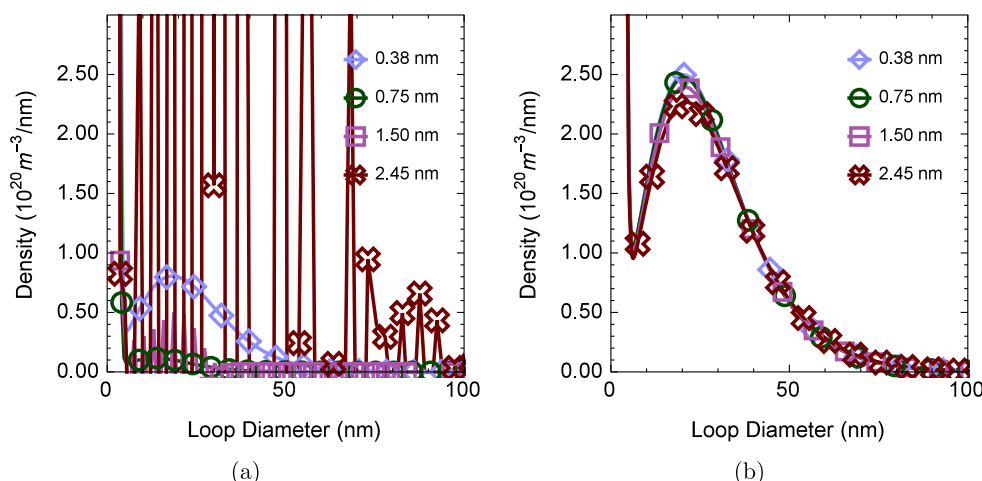


Figure 5. Loop size distributions on a linear scale at 10 dpa for interpolation grouping schemes starting at (a) i_{35} or (b) i_{700} for four group widths. Occasional markers are added to distinguish curves.

Corruption of the discrete distribution due to the cutoff size worsens with time, and by the terminal dose the ungrouped simulations are of no practical utility. The moment method continued to perform well on this problem up to higher doses, with no discernible differences in the final size distribution regardless of group width as shown in figure 4 for four group widths. No noteworthy differences are evident in the distributions on either logarithmic or linear scales as the group width increases, however some oscillations between positive and negative concentrations do appear in domains where the magnitude of the distribution approaches zero. Such oscillations have been reported for other implementations of this scheme previously [20, 21], but as in these cases the magnitude is negligible with a typical magnitude 8–12 orders below the magnitude of the distribution peak.

The solution produced by the interpolation scheme, which had begun to deviate by 1 dpa, was notably poorer by the terminal dose. When the default simulation parameters are used the scheme quickly diverges with group width and in some cases develops significant oscillations as shown in figure 5. At other group widths, where the oscillatory behavior does not develop, the solution still accrues significant error. Unlike the Frenkel pair problem, a clear convergence trend with group width is not immediately evident. We note that for the interpolation scheme the vacancy cluster group width was fixed at a small value in contrast to the moment scheme where group width on the vacancy cluster side was varied along with interstitial group width.

There are several possible explanations for the poor performance of the interpolation scheme on this problem. Before concluding that interpolation simply cannot be used on problems with dense reaction and/or coalescence networks, we extended the analysis of this method slightly. The size distribution that emerges from this problem leaves a high density (near 10^{23}m^{-2}) of small 2–4 nm loops which develop early with a lower density of larger features that develop later. This leaves a sharp elbow in the size distribution.

The starting point of grouping can be changed arbitrarily, so we increased it to a value past the elbow in the SDF. The performance of the interpolative scheme in this case was considerably improved, and the later starting point is shown in figure 5(b). The oscillations present in the original simulations vanished and convergence to the correct SDF emerged.

Key observable features such as loop density and the mode of the SDF at the terminal dose are in agreement between the moment and interpolation scheme, as can be seen by comparing figures 4(b) and 5(b). However, the later starting point incurs significant additional cost as will be discussed later. As with the Frenkel pair problem, the primary error observed as the group size increased was a slight spreading effect on the size distribution, with a slightly lower peak height but similar net density and mean size. Another similarity was that the interpolation scheme began displaying such errors at somewhat smaller group sizes than the moment scheme. The consistency in behavior across these two very different problems merits some discussion.

It has been hypothesized that exact conservation of mass and concentration is fundamental to constructing a functional grouping scheme [21]. For the interpolation schemes presented here, the time derivative of concentration and mass at each node is an ‘exact’ calculation from the assumed SDF in principle, but the time derivative of values between the nodes are—like the concentrations themselves—approximated by interpolation introducing the possibility of mass leakage. In the Frenkel pair case, most of the concentrations necessary to evaluate the reaction network are found at or near nodes. The fraction of terms in the problem which require interpolation is rather small. For the terms which do require interpolation, adjacency to nodes leaves the associated error at a minimum, with values that closely resemble those of the true SDF if every point were tracked explicitly.

For the loop problem, all points in the SDF are required to evaluate the reaction network. Not only does this introduce additional deviation from the ‘true’ SDF from interpolation error, it introduces additional possibility for mass loss in each integration step. Notably, the stiff nature of reaction diffusion problems demands an implicit integration scheme. Consequently, concentration updates at the nodes are dependent on the non-conservative updates to intermediate points as inferred from interpolation. In practice then, the ‘exact’ time derivatives at the nodes are in fact potentially highly non-conservative, and more so as the number of contributions from interpolated points grows. From this perspective, the convergence of the loop problem raises doubts about the necessity of intrinsic conservation. Indeed, even in the moment scheme where mass is conserved to machine precision, the SDF becomes distorted and the cluster sizes and densities diverge from the correct solution eventually as group widths become too wide to fully resolve the cluster distribution.

This all suggests that the conservation issue is less crucial than previous analysis has indicated. Once acceptably converged, the interpolation schemes demonstrate conservation of mass to reasonable tolerance and produce minimal errors in critical observable quantities such as aggregate number densities and mean sizes—even though they are not analytically constructed to do so. Instead, it appears that the assumed SDF must only be a sufficiently close approximation to the discrete SDF to allow the proper physics to operate. Examining the Frenkel pair problem we concluded that the approximation must contain (at a minimum) information relevant to second derivatives. The difference in performance between the interpolation scheme initiated before and after the elbow in the SDF for the loop problem hints that for interpolation schemes, the SDF must be well enough behaved within each group to be described reasonably accurately by the approximating function. Put in other terms, the SDF discretization must be narrow enough to resolve the distribution’s features. The moment scheme will eventually fail by the same criterion as group widths become too wide, but it appears generally more robust.

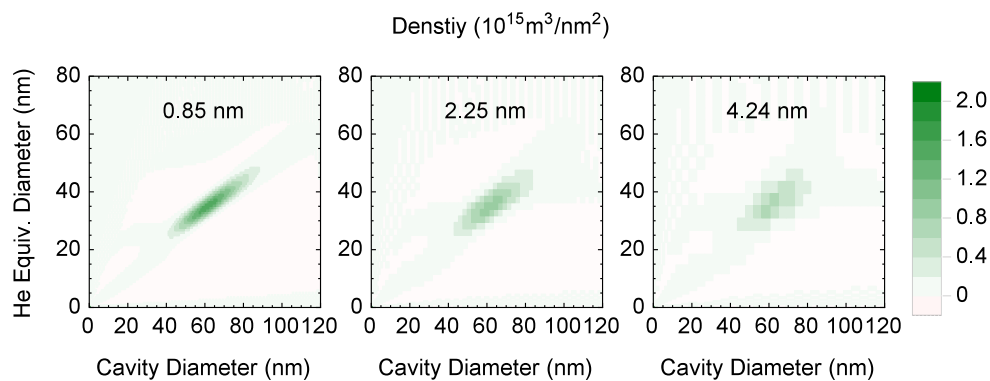


Figure 6. Bubble size distributions for the moment scheme at 100 dpa for group widths of (left) 0.85 nm, (center) 2.25 nm, and (right) 4.24 nm. These distributions are generated based on only the zero moments in order to show the group boundaries clearly.

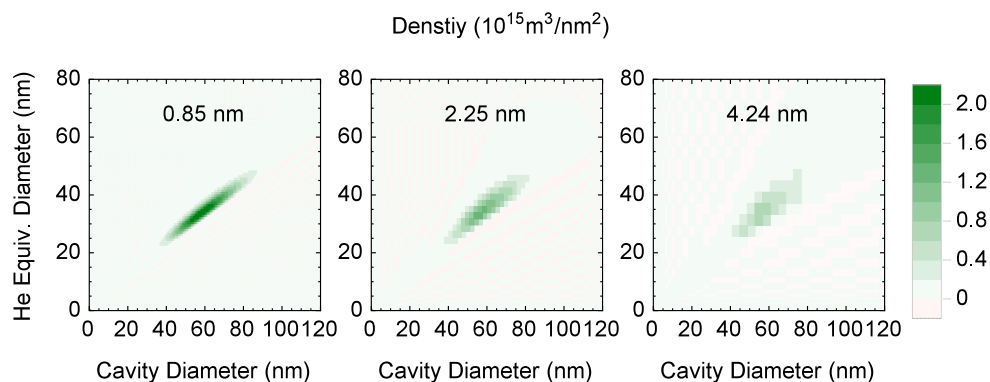


Figure 7. Bubble size distributions for the interpolation scheme at 100 dpa for group widths of (left) 0.85 nm, (center) 2.83 nm, and (right) 4.24 nm. These distributions are generated based on only the zero moments in order to show the group boundaries clearly.

3.4. Bubble problem

The bubble problem tests the ability of each scheme to accommodate multi-component clusters in the model. As with the previous two tests, the moment approach demonstrates excellent performance on these problems. Figure 6 shows the bubble size distributions at the terminal dose for the moment method for three different group widths. These distributions are displayed in terms of the diameter from equation (21) and the equivalent diameter, which is calculated by substituting the number of vacancies in equation (21) with the number of helium. Despite significant loss of resolution for the largest width shown, the size distribution retains all of its most important properties. These include the location of the SDF peak in both the helium and vacancy directions, the net cluster density, and the shape of the SDF (within the resolution limit of the group width).

The low magnitude oscillations between negative and positive values are also apparent in this problem in certain regions of phase space. In other regions, the concentration takes

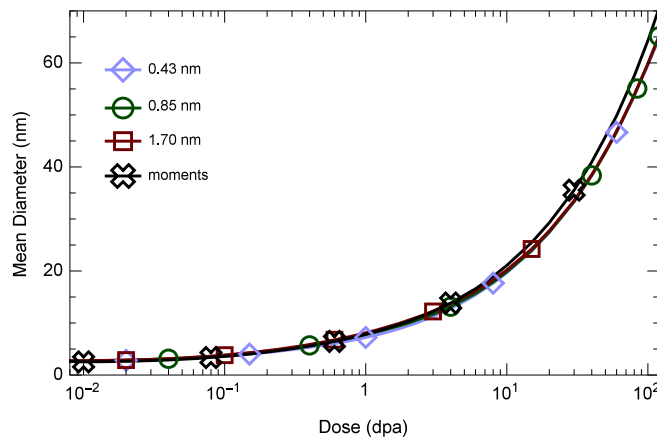


Figure 8. Mean bubble size versus dose for several group widths in the interpolative scheme as compared to the moment scheme.

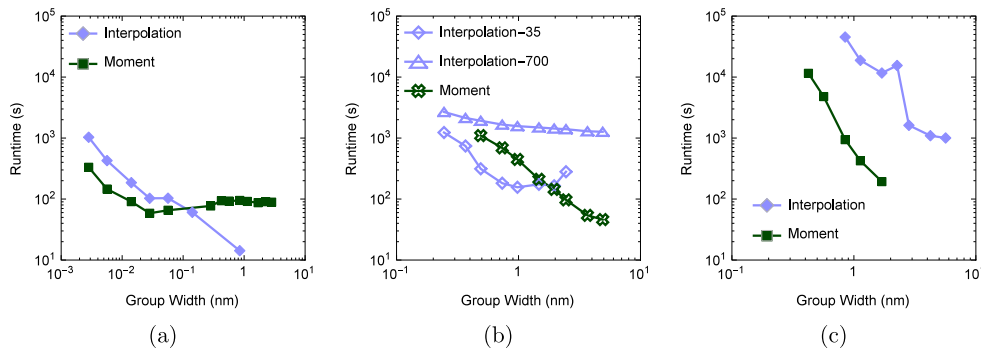


Figure 9. Total run time as a function of group width for the (a) Frenkel pair, (b) loop, and (c) bubble test problems comparing the interpolation and moment grouping schemes. For the loop problem both the i_{35} and i_{700} starting points are included for the interpolation scheme.

consistently negative values. As with the other test problems, the magnitude of any negative values are many orders of magnitude below that of the SDF peak.

The interpolation scheme shows nearly identical performance on these problems. For every group width tested, the simulations reached the terminal dose of 100 dpa without obvious signs of failure. The bubble distributions at the terminal dose for this scheme are shown in figure 7. The behavior of this scheme, and the distributions it produces, are similar but not identical to the moment scheme. Small negative values are also produced by this scheme, but their magnitude is somewhat larger than the moment case. This contrasts with the loop problem where oscillations were more pronounced for the moment scheme. Additionally, the regions of phase space which displayed consistently negative values in the moment scheme display oscillation when interpolation is used.

Figure 8 shows the evolution of the mean bubble size with dose as compared to the moment method. The interpolation method is clearly well converged for the group widths used here, though at slightly different values than the moment method. It is unclear whether

this discrepancy is a product of the schemes themselves or a slight variation in parametrization between the two codes. Regardless, the difference is quite small and both schemes demonstrate stable and consistent solutions across the entire phase space and dose range.

Finally, we turn our attention to the efficiency of these grouping schemes. It is pertinent to mention a few details involved in solving these problems that have not yet been discussed, primarily the issue of phase space boundaries. A hard bound on the SDF must be placed somewhere, and should the solution reach it, the behaviors of both schemes degrade. For the moment scheme, time steps become difficult to solve and mass accumulation at the boundary drives oscillation and in some cases artificial nucleation of new clusters. For the interpolation scheme, mass leaks across the phase space boundary and the cluster distribution eventually grows into non-existence. Both behaviors are obviously undesirable and we enforce an expansion of phase space should any significant fraction of the distribution approaches the boundary. The heuristics governing phase space size influence the cost of the calculations, and the aggression required in such expansions varies from scheme to scheme.

The net run time of each of the test problems is shown as a function of group width in figure 9. The small sizes shown for the Frenkel pair problem were not accessible to the other problems due to memory limitations. Despite the larger memory associated with storing the reaction networks arising from the more numerous interaction coefficients associated with the moment scheme, in some cases it offered shorter total run times than the interpolation scheme for the same group widths. This is clearest in the bubble problem. Some of the extra cost incurred by the interpolation schemes may come from earlier expansions of the phase space, but a more substantial portion originates in the implicit solver, which in the case of the bubble problem required a more expensive preconditioning strategy in the interpolation approach than that of the moment method. It should be noted that in all cases, analytic expressions were used as opposed to direct summation when computing the grouped reaction coefficients following the procedure discussed in appendix A. When attempting to use direct summation, the ‘set up time’ associated with computing these coefficients becomes prohibitive, particularly when applying the moment scheme for problems with dense reaction networks (i.e. the loop problem).

The relationship between cost and group width was of higher order for the loop (figure 9(b)) and bubble (figure 9(c)) problems than the much simpler Frenkel pair (figure 9(a)) problem. The number of components in the reaction network for those two problems decrease with the square of the group width. This is a consequence of the dense reaction network in the loop problem and the two-dimensional phase space in the bubble problem. The cost decreased smoothly with group size in the moment approach, but was somewhat less predictable for the interpolation schemes. As group size continues to increase, returns on computational speed eventually diminish. This can occur when the cost of the problem becomes dominated by the non-grouped region of phase space. For the loop problem with the non-grouped region extended up to the 700th cluster, this occurs below the group size range covered by our simulation matrix.

Both schemes allow simulation of complex problems up to high damage levels at very reasonable cost, whereas a non-grouped solution is clearly unattainable for either the loop or bubble test problem for a single processor code. Without grouping, the discrete problems may be solvable by brute force with parallel computing, though this route is unnecessarily expensive in our view (and likely still unfeasible for multi-component simulations). Further, utilizing a grouping scheme allows the possibility of using the parallel environment to apply the cluster dynamics approach in a large-scale, spatially-dependent manner.

Generally, both the two moment approach and interpolation approach to constructing groups appear viable and capable of producing reliable solutions. These methods demonstrate

similar behavior on all three of the test problems considered in this work, and reproduce the solution from discrete cluster dynamics where it is available. Both approaches are capable of handling multiple component axes in phase space and coalescence between larger features in the SDF. However, both exhibit a number of weaknesses that should be carefully considered. Foremost among these is the production of oscillations in the size distribution which can potentially produce non-physical or divergent solutions. Broadly speaking, the moment approach is more stable against such behavior, but both are ultimately vulnerable and group widths need to be chosen such that they clearly resolve any features in the SDF. Further, care must be taken to ensure that the SDF has not reached the phase space boundary, as both schemes behave unpredictably in such circumstances.

4. Conclusions

This paper has laid out a framework to reduce the computational cost of the cluster dynamics approach to simulating complex reaction diffusion processes by grouping size distributions to reduce the number of equations. Using this framework, we have developed several schemes and analyzed their performance on three test problems. This includes the generalization of an existing scheme designed for monomer transport to arbitrarily complex problems and the introduction of a class of novel schemes. The former, which follows multiple moments of the SDF, demonstrated excellent stability and fast convergence in each case. The novel schemes, based on interpolating the SDF, required interpolating functions of at least second order to yield convergence to the discrete SDF.

Three test problems designed to target the physics typically involved in radiation damage were used to evaluate the viability of both the interpolation and moment schemes. Both approaches were capable of generating convergent solutions for the density and size evolution of defect cluster populations in all three problems. Both approaches are clearly capable of handling the third order kinetics present in the loop problem, the dense reaction networks present in the loop problem, and the multi-component cluster phase spaces present in the bubble problem. All three features, namely the complex reaction networks, multi-component systems, and one-dimensionally migrating clusters, are likely to be necessary for high fidelity modeling of radiation induced changes in material microstructures.

The interpolation schemes are significantly easier to implement and may eventually prove more efficient with further algorithmic or computational development. Unfortunately, at present their performance varied significantly compared to the moment scheme, both in terms of computational cost and ability to produce a well-converged SDF. For use on general problems, interpolation requires some procedure to choose group sizes and the grouped range to avoid such instabilities, and there is no obvious means of doing so *a priori*. Alternatively, one might enforce mass conservative box constraints to prevent negative concentrations from developing, which can be implemented at reasonable cost [34]. Without such a safeguard, we recommend use of the moment approach for general cluster dynamics problems due to its superior robustness and more predictable performance. Both approaches, however, allow the efficient application of cluster dynamics methods to problems which would otherwise require as many as 10^{18} coupled equations.

Acknowledgments

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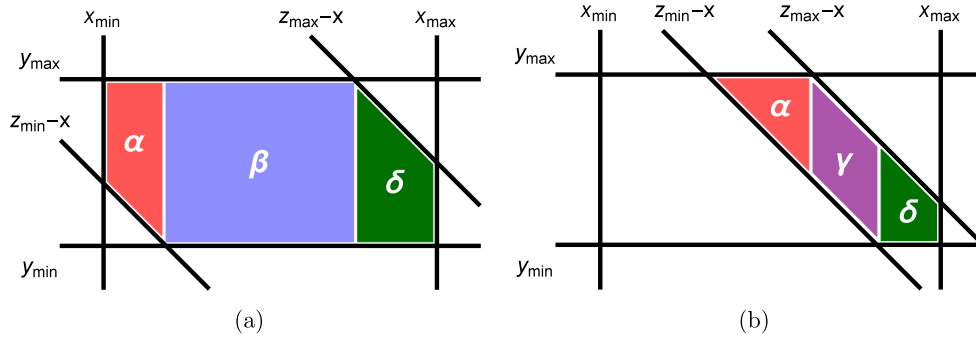


Figure A1. Illustration of group reaction boundary conditions for group the groups x and y reacting to form group z .

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Appendix A. Analytic group formulation

In order to efficiently construct the grouped reaction coefficients, the summations inherent in coefficient equations such as equations (7a)–(7d), (11) and (12) should not be evaluated through direct summation, but rather by analytical expressions which compute them as a function of the group widths and locations. Arriving at such expressions is not entirely trivial. The bounds on the sums are quite complicated, particularly for the moment scheme, where a double summation is required. Further, the replacement of the discrete sums with integration can compromise any desirable conservation properties, and degrade the ultimate accuracy of the schemes. Fortunately the advent of computer algebra systems render the formulation of such expressions, however complex, quite possible. In this section we illustrate this process where it is most complex, for the moment scheme.

In this case, posing the correct bounds on the sums is key. In figure A1 we illustrate the four possible boundary conditions imposed on a general reaction between two parent groups (i and j) to produce a single daughter group (k). For the notation in this section we refer to the domain of the parent groups as x and y and the daughter group as z . If we consider summing externally over the clusters in x and internally over the clusters in y , there are four possible boundaries for the inner sum.

One condition, herein noted α , occurs when the lower bound is governed by the daughter group and the upper bound by the parent group y . The second condition β occurs when both boundaries are governed by the limits of the parent group. The third condition γ occurs when both bounds are determined by the daughter group. Finally, the δ condition occurs when the upper summation bound is governed by the daughter group, but the lower bound is determined by the parent group. Up to three of these conditions can be present in the same coefficient equation (β and γ cannot coexist), and as few as zero.

For each set of conditions, a unique problem emerges. The bounds on each condition in the domain of group x can be defined as follows:

$$\alpha_{\min} = \max(x_{\min}, z_{\min} - y_{\max}), \quad (\text{A.1})$$

$$\alpha_{\max} = \min(x_{\max}, z_{\min} - y_{\min} - 1, z_{\max} - y_{\max} - 1), \quad (\text{A.2})$$

$$\beta_{\min} = \max(x_{\min}, z_{\min} - y_{\min}), \quad (\text{A.3})$$

$$\beta_{\max} = \min(x_{\max}, z_{\max} - y_{\max}), \quad (\text{A.4})$$

$$\gamma_{\min} = \max(x_{\min}, z_{\max} - y_{\max}), \quad (\text{A.5})$$

$$\gamma_{\max} = \min(x_{\max}, z_{\min} - y_{\min}), \quad (\text{A.6})$$

$$\delta_{\min} = \max(x_{\min}, z_{\max} - y_{\max} + 1, z_{\min} - y_{\min} + 1), \quad (\text{A.7})$$

$$\delta_{\max} = \min(x_{\max}, z_{\max} - y_{\min}) \quad (\text{A.8})$$

and the summation over the entire reaction space can be decomposed as

$$\alpha_{i,j,k}^{r,s,t} = \sum_{x=\alpha_{\min}}^{\alpha_{\max}} \sum_{y=z_{\min}-x}^{y_{\max}} v_i^r(x) \times v_j^s(y) \times \theta_k^t(x+y), \quad (\text{A.9})$$

$$\beta_{i,j,k}^{r,s,t} = \sum_{x=\beta_{\min}}^{\beta_{\max}} \sum_{y=y_{\min}}^{y_{\max}} v_i^r(x) \times v_j^s(y) \times \theta_k^t(x+y), \quad (\text{A.10})$$

$$\gamma_{i,j,k}^{r,s,t} = \sum_{x=\gamma_{\min}}^{\gamma_{\max}} \sum_{y=z_{\min}-x}^{z_{\max}-x} v_i^r(x) \times v_j^s(y) \times \theta_k^t(x+y), \quad (\text{A.11})$$

$$\delta_{i,j,k}^{r,s,t} = \sum_{x=\delta_{\min}}^{\delta_{\max}} \sum_{y=y_{\min}}^{z_{\max}-x} v_i^r(x) \times v_j^s(y) \times \theta_k^t(x+y) \quad (\text{A.12})$$

with a , the total coefficient, constructed as

$$a_{i,j,k}^{r,s,t} = k_{i,j}(\alpha_{i,j,k}^{r,s,t} + \beta_{i,j,k}^{r,s,t} + \gamma_{i,j,k}^{r,s,t} + \delta_{i,j,k}^{r,s,t}), \quad (\text{A.13})$$

where r , s , and t refer to the moments of the two parent groups and the daughter group. In these equations there is a function, θ which arises from equations (17) and (18) which is defined as

$$\theta_k^t(z) = \begin{cases} \frac{1}{(1+2\Delta x_k)} & t = 0 \\ \frac{3(z-x_k^0)}{\Delta x_k(1+2\Delta x_k)} & t = 1 \end{cases} \quad (\text{A.14})$$

over the product group (where Δx_k is the width parameter of group k and should not be confused with the parent domain variable x). Each of these component summations can be replaced with an analytic function of the bounds on each group and then the total coefficient computed as the sum of each piece

$$a_{i,j,k}^{r,s,t+} = k_{i,j}^+(\alpha_{i,j,k}^{r,s,t} + \beta_{i,j,k}^{r,s,t} + \gamma_{i,j,k}^{r,s,t} + \delta_{i,j,k}^{r,s,t}). \quad (\text{A.15})$$

Nominally, the above process is formulated for calculating reaction gains (the a^+ coefficients in equations (20a) and (20b)). However, it can easily be applied to dissociation.

For multi-component systems, these equations can be applied independently using the limits along each component axis. The product of equation (A.15) for each component gives the full coefficient. Further, the reaction loss corresponding to each gain can be calculated by replacing the θ for the daughter group with one for a parent group. If the products of the two

Table B1. Cluster properties for the Frenkel pair test problem.

Cluster	D_0 (nm ² s ⁻¹)	E^m (eV)	N	Geometry	Emission
i_1	2×10^{11}	0.34	3	Sphere	None
v_1	2×10^{11}	0.67	3	Sphere	None
$v_{2 \rightarrow \infty}$	0	N/A	N/A	Sphere	None

Table B2. Cluster properties for the loop test problem.

Cluster	D_0 (nm ² s ⁻¹)	E^m (eV)	N	Geometry	Emission
i_1	2×10^{11}	0.34	3	Sphere	None
i_2	$D_{0,i1}/2$	0.42	3	Sphere	i_1
i_3	$D_{0,i1}/3$	0.43	3	Sphere	i_1
$i_{4 \rightarrow \infty}$ —free	$D_{0,i1} \times n^{-0.7}$	0.1	1	Loop	i_1
$i_{4 \rightarrow \infty}$ —trapped	0	N/A	N/A	Sphere	i_1, trap
v_1	2×10^{11}	0.67	3	Sphere	None
v_2	$D_{0,v1}/2$	0.62	3	Sphere	v_1
v_3	$D_{0,v1}/3$	0.37	3	Sphere	v_1
v_4	$D_{0,v1}/4$	0.48	3	Sphere	v_1
$v_{5 \rightarrow \infty}$	0	N/A	N/A	Sphere	v_1

Table B3. Cluster properties for the bubble test problem.

Cluster	D_0 (nm ² s ⁻¹)	E_m (eV)	N	Geometry	Emission
i_1	2×10^{11}	0.34	3	Sphere	None
v_1	2×10^{11}	0.67	3	Sphere	None
v_2	$D_{0,v1}/2$	0.62	3	Sphere	v_1
v_3	$D_{0,v1}/3$	0.37	3	Sphere	v_1
v_4	$D_{0,v1}/4$	0.48	3	Sphere	v_1
$v_{5 \rightarrow \infty}$	0	N/A	N/A	Sphere	v_1
he_1	2×10^{11}	0.06	3	Sphere	None
$he_{2 \rightarrow 3}$	$2 \times 10^{11}/n$	0.06	3	Sphere	i_1, he_1
$he_{4 \rightarrow 8}$	0	N/A	N/A	Sphere	i_1, he_1
$v_{1 \rightarrow \infty} he_{1 \rightarrow \infty}$	0	N/A	N/A	Sphere	i_1, v_1

reacting parents are spread over multiple daughter groups, the equations can be applied separately for each possible daughter and the rates summed.

The number of terms in the analytic equations can vary from just one for some of the β equations to several dozen for some of the α and δ sums. Despite the occasional complexity of these expressions, using them is significantly less expensive than computing the sums directly. A full list of the analytical expressions can be found in the supplementary material.

Appendix B. Parameter catalog

The diffusion and configuration properties used for the three test problems are given in tables B1, B2, and B3. The supplied properties include the diffusion prefactors (D_0),

migration energies (E^m), diffusion dimensionality (N), the assumed geometric configuration of the defects in the cluster, and which monomers (if any) are allowed to dissociate from the cluster. Small cluster migration energies were selected from *ab initio* calculations [35, 36] while the behavior of larger clusters was inferred from experiment as discussed in the main text. The diffusion coefficient is computed in the usual way

$$D_i = D_{0,i} \times \exp \frac{E_i^m}{k_b T} \quad (\text{B.1})$$

and the discrete reaction rate coefficients are computed as

$$k_{i,j}^+ = 4\pi (D_i + D_j)(r_i + r_j + r_0) \quad (\text{B.2})$$

for the spherical clusters. The r_0 parameter is applied equivalently to all interactions at a value of $a_0 \times 3\sqrt{3}/4$ where the lattice parameter a_0 was taken to be 0.287 nm. The radii for a spherical defect cluster in the bcc lattice is defined as

$$r_i = a_0 \left(\frac{3n}{8\pi} \right)^{1/3}, \quad (\text{B.3})$$

where n is the number of defects in the cluster. The reaction rates are different for the non-spherical clusters and one-dimensional diffusers in the loop problem, and the means of treating these clusters is available elsewhere [4, 30].

The dissociation rates are computed using

$$k_{i,j}^- = \frac{k_{i-j,j}^+}{\Omega} \exp \frac{-E_{i,j}^b}{k_b T} \quad (\text{B.4})$$

with the atomic volume Ω equivalent to $a_0^3/2$. The binding energies in this expression were based on capillarity laws

$$E_{v*,v}^b = 1.73 - 2.59(n^{2/3} - (n-1)^{2/3}), \quad (\text{B.5a})$$

$$E_{i*,i}^b = 4.33 - 5.76(n^{2/3} - (n-1)^{2/3}), \quad (\text{B.5b})$$

$$E_{v*,he*,v}^b = 1.73 - 2.59(n^{2/3} - (n-1)^{2/3}) + 2.50 \log(1 + n_{he}/n_v), \text{ and} \quad (\text{B.5c})$$

$$E_{v*,he*,i}^b = 4.88 + 2.59(n^{2/3} - (n-1)^{2/3}) - 2.50 \log(1 + n_{he}/n_v), \quad (\text{B.5d})$$

where the latter two expressions were designed as minimal parameter fits to the atomistic data [37].

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